

Do LLM Fingerprints Survive Outside English?

Interpretable Attribution of Five Frontier Models Across Seven Languages

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Abstract—Tools that attribute AI-generated text to the model that wrote it are built and evaluated mostly in English, while the settings where attribution matters, such as disinformation tracing and content moderation, are multilingual. We ask whether the stylistic fingerprint of a specific large language model (LLM) survives outside English, and whether an interpretable representation can carry it. We collected 840 persuasive essays from five frontier models (GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3, DeepSeek V4 Pro) in seven languages ordered by training-data resource level, holding content fixed with natively authored prompts, and described each text by 21 features defined on Universal Dependencies so that every feature has the same meaning in every language. A logistic-regression classifier attributes texts five ways at .59–.72 accuracy in every language (chance 20%, leave-one-prompt-out), and accuracy does not decline with resource level: the fitted gradient is -0.038 log-odds per rank step (95% CI -0.12 to +0.04). A character n -gram classifier is 3–22 points more accurate within a language, but the two representations part ways at language boundaries: after within-language standardization the interpretable features transfer between languages at mean accuracy .49 against .41 for n -grams, and a classifier trained on English reads machine translations of English text at .664 against .29, because paragraphing, sentence rhythm and punctuation survive translation while vocabulary does not. Models converge stylistically as resources fall (centroid separation vs. rank, $\rho = -0.96$) without becoming indistinguishable, and the five models themselves, asked which model wrote each text, score 20.3%–25.4%. Code, prompts, all texts and translations are released.

Index Terms—authorship attribution, AI-generated text, multilingual text mining, stylometry, large language models

I. INTRODUCTION

Most work on machine-generated text asks whether a text was written by a person or by a model. A second question matters for provenance: *which* model wrote it. Answering it lets an analyst link an influence campaign to a toolchain or apply platform rules that differ by provider. The attribution literature is mostly English [1], [2], [3], and the two multilingual studies we know of use fine-tuned or probability-based classifiers whose decisions cannot be inspected [4].

There are two reasons to doubt that English results carry over. First, models see far less training text in lower-resource languages [5], so their writing there may be less distinctive or less stable. Second, languages differ in script and morphology, so many classic stylometric measures (readability formulas, contractions, hedge lists) are undefined outside English. Character n -grams, the standard strong baseline in authorship at-

tribution [6], [7], have the opposite problem: they are accurate but tied to the script and vocabulary of the language they were fit on.

We address both problems with a design that holds content fixed across languages and describes every text with features defined on Universal Dependencies (UD) [8], and we measure what that choice costs and buys against an n -gram baseline. We ask six pre-registered questions. **RQ1**: Can five frontier models be told apart within each of seven languages? **RQ2**: Does attribution weaken as a language’s resource level falls? **RQ3**: Is a model’s fingerprint shared across languages or specific to each? **RQ4**: Do models converge on one style in lower-resource languages? **RQ5**: Does machine translation erase the fingerprint? **RQ6**: As a baseline, can the models themselves identify the author? A seventh analysis, added in response to review, asks what the interpretable representation gives up relative to n -grams and where it wins.

Our contributions are: (i) a content-controlled corpus of 840 essays (5 models $\times$ 7 languages $\times$ 12 prompts $\times$ 2 generations) and 1,440 machine translations, released with code; (ii) a 21-feature representation with one definition in every language, reaching .59–.72 five-way accuracy in all seven languages with no detectable resource gradient; (iii) a comparison with character n -grams showing that the opaque baseline is more accurate within a language but only the interpretable features carry a classifier across a language or translation boundary without retraining; (iv) evidence that models converge stylistically in lower-resource languages while remaining attributable, and that LLM judges attribute near chance. Decision rules were fixed before data collection; additions are labelled.

II. RELATED WORK

Detection. Zero-shot detectors [9] and multilingual benchmarks [10] separate human from machine text; detectors fine-tuned on English lose about a quarter of their F1 on other languages [11]. These works ask human versus machine, not which machine.

Model attribution. Uchendu et al. framed attribution among neural generators and built TuringBench [1], [2]; Kumarage and Liu applied stylometric features to LLM attribution [12]; Sun et al. showed that embedding classifiers separate current model families and that the signal survives

LLM translation into Chinese [3], all in English or English-centred. La Cava et al. attribute among seven LLMs across 18 languages with fine-tuned and probability-based detectors and report cross-lingual transfer [4]. Stylometric detection has been applied language by language with language-specific feature sets, e.g. Japanese [13] and Polish [14]; UD-based features have been used for cross-language attribution of human authors [15]. To our knowledge, no prior study attributes text to its source model with features that have the same definition in every language, holds content fixed across languages, or tests attribution against a resource gradient; see [16] for a survey.

Homogenization and LLM judges. Writing with LLMs reduces the diversity of human text [17], [18]; whether models converge on *each other* where training data is thin has not been tested. LLM evaluators recognize and favour their own outputs [19]. We use LLM judges only as a baseline; a companion study examines self-recognition in English under two elicitation formats [20].

III. DATA AND METHOD

A. Models, languages and prompts

The five subject models, queried through OpenRouter with the served model string logged for every response, are GPT-5.5, Gemini 3.5 Flash, Claude Opus 4.7, Grok 4.3 and DeepSeek V4 Pro. The seven languages are English, Spanish, Chinese, Russian, Japanese, Turkish and Hindi, assigned ordinal resource ranks 1–7 in that order to reflect their approximate share of web training data [5]; we use a rank because numeric estimates disagree by source.

Twelve English prompts each fix a topic, a stance (six for, six against), three supporting claims, a prose-only instruction and a reading-level tier. For each language a non-subject model (Llama 4 Maverick) wrote a native version from those elements rather than translating the English. A model from a third family (Qwen 3.7 Plus) then read each of the 84 prompts blind, extracted the topic, stance and claims, and compared them with the original; prompts that failed were regenerated with the reviewer’s note until all 84 passed.

B. Generation, validation and features

Each prompt was sent as a single user turn, no system prompt, temperature 0.7, twice per cell with fixed seeds and reasoning traces excluded: $5 \times 12 \times 7 \times 2 = 840$ responses, each passing language identification, a 150–900 word-equivalent window and a refusal check. One Grok response to a Hindi prompt was written in English and excluded, leaving 839 texts; ten DeepSeek responses that spent a 4,000-token budget on hidden reasoning were re-collected at 16,000.

Every text was parsed with Stanza [21] in all seven languages, with Chinese and Japanese pre-split on sentence-final punctuation. We compute 21 features in five groups. *Lexical*: moving-average type–token ratio (MATTR, window 50) [22], hapax rate, mean token length, Zipf slope. *Syntactic*: mean and SD of sentence length, burstiness [23], mean maximum dependency depth, subordinate-clause rate (acl, advcl, ccomp,

xcomp, csubj per sentence), function-word ratio, first-person rate. *Structure*: paragraph count and length, question rate, connective rate. *Punctuation*: commas, colons, dashes and semicolons per 1,000 tokens, with full-width and script variants mapped to the same feature. *Character*: character-bigram entropy and digit rate. English-only measures such as readability formulas, contractions and hedges are excluded.

C. Analysis

The unit of analysis is one response; the two generations of a prompt are not independent, so every split is leave-one-prompt-out (LOPO, 12 folds) and every interval and clustered standard error groups by prompt. Tests against chance are one-sided exact binomials; comparisons between classifiers on the same texts are exact McNemar tests; the 28 headline comparisons (features vs. chance, n -grams vs. chance, features vs. n -grams, features vs. the best LLM judge, in seven languages) are Holm-corrected as one family.

RQ1. Within each language, multinomial logistic regression on standardized features (primary, interpretable) and a random forest (ceiling), with prompt-clustered bootstrap intervals. **Baseline.** TF-IDF over character 1–4-grams fit inside each training fold, then logistic regression; word 1–2-grams likewise. **RQ2.** A binomial GLM of cell-level correctness on resource rank with clustered standard errors (by prompt, by language, and by both), plus Spearman’s ρ over the seven per-language accuracies; as a robustness check, features residualized on log length within each language. **RQ3.** For each feature, a two-way ANOVA (model $\times$ language, prompt as a blocking factor) with partial η^2 ; and a transfer matrix in which features are z-scored within each language, a classifier is trained on language A and tested on B . For n -grams the fair counterpart fits the vocabulary on the union of A and B (no labels) and z-scores TF-IDF columns within each language, the same unlabelled adaptation. **RQ4.** After z-scoring within each language, the mean pairwise distance between model centroids, the silhouette of model labels and the between/within scatter ratio, with stratified bootstrap intervals and Spearman’s ρ against rank. **RQ5.** All 120 English responses were translated into the six other languages by Google Translate and by a free open model that is not a subject (MiniMax M3). We measure attribution on the translations alone, transfer from classifiers trained on the English originals or on native target-language responses, and per-feature Spearman correlation between original and translation. **RQ6.** Each subject model was shown every text without metadata and asked which of the five models wrote it, one text at a time with a candidate list, in the single-text protocol of [24]: option order shuffled, temperature 0, hidden reasoning disabled where the API allows it. **Feature economy.** A nested feature-selection curve: within each LOPO fold, features are ranked by mean absolute standardized coefficient on the training fold only, the top k are refit, and the held-out prompt is scored, for $k = 1, \dots, 21$; and a per-language ablation of each feature group.

TABLE I

PER-LANGUAGE RESULTS. LR: LOGISTIC REGRESSION ON THE 21 UD FEATURES, ACCURACY [PROMPT-CLUSTERED 95% CI]; F1: MACRO-F1; n -GRAM: CHARACTER 1–4-GRAM TF-IDF [CI]; JUDGES: MEAN OF THE FIVE LLM JUDGES (BEST). CHANCE ACCURACY IS 0.20.

Language	Rank	LR [95% CI]	F1	n -gram [CI]	Judges
English	1	.717 [.61–.82]	.711	.750 [.71–.79]	.230 (.267)
Spanish	2	.625 [.53–.70]	.625	.733 [.65–.80]	.207 (.217)
Chinese	3	.650 [.55–.73]	.644	.867 [.81–.93]	.208 (.233)
Russian	4	.650 [.52–.78]	.654	.783 [.66–.88]	.237 (.258)
Japanese	5	.708 [.63–.78]	.707	.792 [.71–.87]	.198 (.225)
Turkish	6	.592 [.50–.69]	.596	.783 [.73–.84]	.205 (.258)
Hindi	7	.639 [.56–.71]	.634	.782 [.73–.83]	.247 (.361)

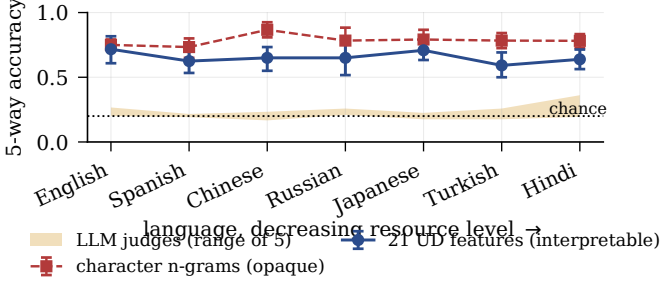


Fig. 1. Five-way attribution accuracy by language (LOPO, prompt-clustered 95% CIs). The opaque n -gram baseline is more accurate within every language; neither representation declines with resource level; the five LLM judges stay near chance.

IV. RESULTS

A. RQ1: Attribution works in every language

Table I and Fig. 1 show five-way accuracy of .59–.72 in every language; every prompt-clustered interval excludes chance and every Holm-corrected binomial has $p < 10^{-4}$. Per-model recall is highest for GPT-5.5 (67%–92%) and Grok 4.3 (63%–88%); DeepSeek V4 Pro is hardest (25%–58%). Length is part of the signal: Grok writes the shortest responses in all seven languages. With features residualized on length, logistic regression falls to .44–.57 while the random forest, which the plan reserves as a ceiling, stays at .66–.78; we read this, exploratorily, as a length-independent but non-linear fingerprint.

The baseline is more accurate. Character n -grams reach .73–.87 (Table I), 3–22 points above the features; word n -grams average .74. After Holm correction the n -gram advantage is significant in 2 of seven languages (Chinese, Turkish) and not in the others. This is expected [7]: n -grams see vocabulary. What the features give up within a language they recover across languages (Sec. IV-C) and under translation (Sec. IV-E).

Feature economy. By mean absolute coefficient the fingerprint is carried mainly by paragraph count, comma rate, char-bigram entropy, paragraph length, Zipf slope. The nested selection curve (Fig. 2) shows five features already reach macro-F1 .50–.75 and ten come within 9 points of all 21 in every language. No feature group is necessary: dropping

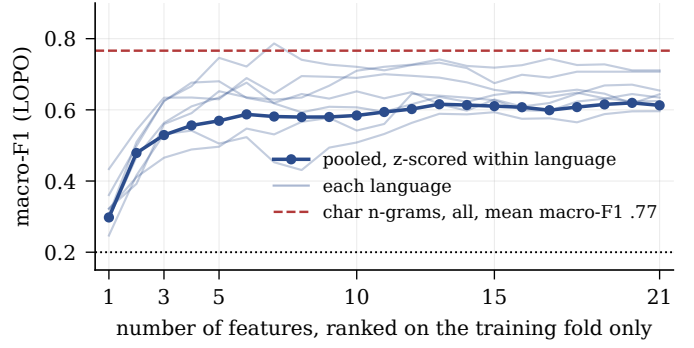


Fig. 2. Nested feature-selection curve. Features are ranked on the training fold only; five features carry most of the signal.

any one costs at most .11 macro-F1, and structure alone gives .31–.61, punctuation alone .26–.53. A character model that sees only punctuation, digits and whitespace, with every word replaced by a placeholder, attributes at .62 within language, direct evidence that the fingerprint is structural.

B. RQ2: No detectable resource gradient

The cell-level GLM gives $\beta = -0.038$ log-odds per rank step (SE 0.042, 95% CI -0.12 to +0.04, $p = 0.361$ clustered by prompt; $p = 0.226$ clustered by language; $p = 0.343$ by both). The interval is compatible with a decline of up to 17 accuracy points from English to Hindi, so the result is a null with bounded power, not a demonstration of invariance. The point estimate is under one accuracy point per step, and the ordering is not monotone: Japanese (rank 5) matches English and Hindi (rank 7) beats Spanish and Turkish. With length-residualized features $\beta = -0.053$ ($p = 0.164$); Spearman’s ρ over the seven languages is -0.45 ($p = 0.310$). The n -gram baseline shows no gradient either: Hindi (.782) exceeds English (.750), $\rho = 0.32$. Our pre-specified criterion for a gradient (negative coefficient, $p < 0.05$) is not met by either representation.

C. RQ3: A shared fingerprint with a language accent

Averaged over the 21 features, partial η^2 is .54 for language, .145 for model, .13 for the model $\times$ language interaction and .16 for prompt. Language dominates every feature tied to morphology or vocabulary (token length .99, function-word ratio .94, hapax rate .93). Model effects are largest for hapax rate (.38), Zipf slope (.37), comma rate (.32) and paragraph count (.29), and smallest for subordination, question rate and dependency depth ($\leq .03$): the models differ in vocabulary variety, punctuation and paragraphing, not in clause structure.

Removing each language’s mean exposes the shared part (Fig. 3, left). A classifier trained on any language reaches .37–.59 on any other (mean .49, against .65 within language), and all 42 off-diagonal cells exceed chance after Holm correction. About 74% of within-language accuracy transfers. Given the same unlabelled adaptation, the n -gram classifier transfers at mean .41 (Fig. 3, right), with 41 of 42 cells above chance; without it, with vocabulary from the training language only, it

transfers at .28, and 14 of the 42 cells are constant predictions because no character sequence is shared across scripts. The features are not the only representation that transfers, but they transfer more and need no target-language text to do so. The setting is transductive in one respect: the test language’s own means and SDs (no labels) standardize its features, and all languages share the same 12 prompts.

D. RQ4: Models converge where data is thin

All three separation measures fall with resource rank. Mean centroid distance drops from 3.59 in English to 2.69 in Hindi ($\rho = -0.96$, $p = 4.5e - 04$) and the Hindi interval lies below the English one; the between/within scatter ratio falls from .41 to .18 ($\rho = -1.00$) with intervals also separated; the silhouette falls from 0.04 to -0.02 ($\rho = -0.93$, $p = 0.003$), a change within noise of zero, so the pre-registered rule (all three intervals separated) is met for two measures of three. Pairwise distances locate the change: the largest English gaps, GPT–Grok (5.69), Claude–Grok (5.42) and DeepSeek–Grok (4.86), shrink to 2.69, 3.46 and 2.02 in Hindi. Grok’s terse English register is the outlier that disappears; the other four were never far apart. Convergence and stable attribution are compatible: centroid distance averages over all 21 features, most of which converge, while a classifier needs only the few that still separate the models.

E. RQ5: Translation does not launder the fingerprint

Attribution on translated text alone runs .57–.70 (Table II), every interval excluding chance; translating costs about nine points relative to the English originals (.717), about what moving to a native lower-resource language costs. A classifier trained on the English originals reads the translations at .664 (Google) and .660 (LLM), one trained on native target-language responses at only .521–.531: the translator carries over the source’s structural habits. Per-feature survival shows why: paragraph count ($\rho = 1.00$), paragraph length (.98), sentence length (.89) and burstiness (.87) pass through translation almost unchanged, while MATTR (.32), function-word ratio (.46) and bigram entropy (.49) are rewritten by the target language. The two translators behave almost identically.

n -grams retrained on the translations attribute them well (.63–.78), but an English-trained n -gram model applied across the boundary collapses to .21–.24 on every non-Latin script (.57 on Spanish, which shares the alphabet). Only the interpretable features carry a classifier across a translation boundary without retraining. This extends the English-to-Chinese result of [3] to six languages, two translators and a representation whose surviving components can be named.

F. RQ6: LLM judges are near chance

Asked one text at a time which of the five models wrote it, the judges score 20.3% (DeepSeek) to 25.4% (Claude). Only Claude is above chance ($p < 10^{-4}$), and only Claude names its own text more often than others’ (own-text recall 17% vs. 7% false self-claims; Fisher $p < 10^{-4}$), the registered prediction. Each judge defaults to one answer: GPT-5.5 names itself for

TABLE II
ATTRIBUTION AFTER MACHINE TRANSLATION OF THE 120 ENGLISH RESPONSES INTO SIX LANGUAGES (LLM: MINIMAX M3). “TRAINED ON” ROWS TEST ON THE TRANSLATIONS. REFERENCE: ENGLISH ORIGINALS .717; NATIVE RESPONSES IN THE SIX LANGUAGES, MEAN .644.

	Google	LLM
<i>UD features</i>		
LOPO on translations, mean	.625	.629
Trained on English originals	.664	.660
Trained on native responses	.521	.531
<i>Character n-grams</i>		
LOPO on translations, mean	.72	
Trained on English originals	.29	

82% of texts regardless of author, and the others mostly name Claude or GPT-5.5. The feature classifier exceeds the best judge by 28–48 points in every language (Holm-corrected $p < 10^{-4}$). This is the single-text, instant-answer condition; a side-by-side lineup with reasoning enabled raises the same judges, and Claude’s self-recognition in particular, well above these figures [20].

V. DISCUSSION

Three practical points follow. First, interpretable attribution does not require English, and a classifier trained in one language transfers to the others. Second, the choice between an opaque and an interpretable representation is a choice about where the classifier will be used: within one language, with training data in that language, character n -grams are better by 3–22 points; across a language or translation boundary, without retraining, only the UD features are usable. Third, machine translation does not hide which model wrote a text, because the fingerprint is structural, and asking a model who wrote a text is no substitute for measurement.

The convergence result bears on the homogenization literature [17], [18]: model-generated text in lower-resource languages is closer to a shared register than in English, and as more text in these languages is machine-written that register may become the default style readers and future training sets encounter.

Dual use. The same pipeline can attribute text to a person, and machine translation offers pseudonymous writers little protection against structural stylometry. We release only model-generated text.

Limitations. (i) One genre, persuasive essays with a three-claim structure; paragraph count partly reflects instruction following. (ii) Length is entangled with style; the length-controlled result holds only non-linearly. (iii) Resource level is an ordinal rank over seven languages, and the RQ2 interval admits a decline of up to 17 points. (iv) Two generations per cell give prompt-clustered intervals of about ± 9 points. (v) Prompt fidelity was checked by a model, not by native speakers. (vi) Parser quality falls with resource rank and is confounded with it. (vii) Results describe model versions served in September 2026, and judges were tested without extended reasoning.

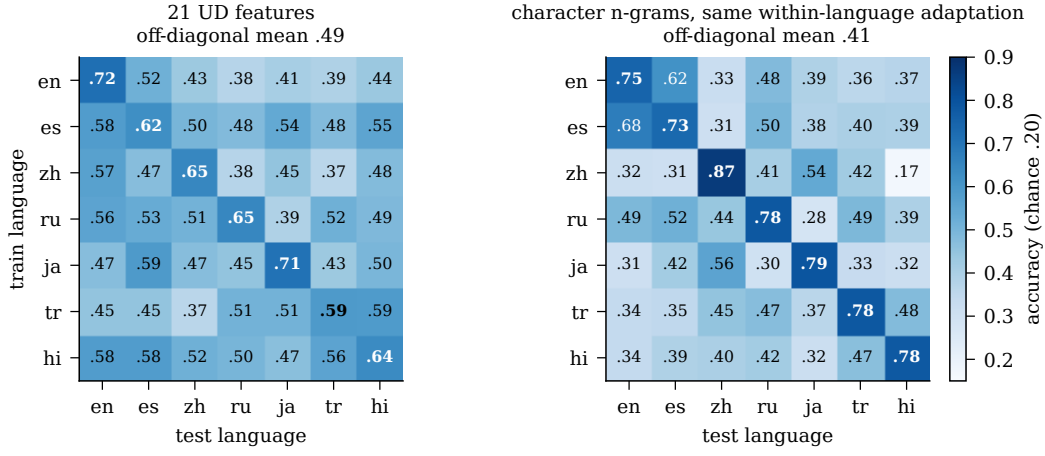


Fig. 3. Cross-lingual transfer. A classifier trained on one language (rows) is tested on another (columns) after within-language z-scoring; diagonal: leave-one-prompt-out within language. Left: UD features. Right: character n -grams given the same adaptation (union vocabulary, z-scored per language).

VI. CONCLUSION

Across seven languages and five frontier models, 21 interpretable UD-based features attribute text to its source model at .59–.72 accuracy with no detectable decline in lower-resource languages. A character n -gram baseline is more accurate within a language and equally flat across the gradient, but only the interpretable features carry a classifier across languages and through machine translation without retraining, because the fingerprint they capture is structural: five features, led by paragraphing and punctuation, carry most of it. The models’ styles converge where training data is thin, and the models themselves cannot read the fingerprint. Future work: other genres, human-verified prompts, numeric resource covariates, reasoning-enabled judges.

Data availability. Code, prompts and reviews, all 839 responses, 1,440 translations, 4195 judge calls, features and result tables: <https://github.com/adrian-erlikhman/LangLLM>.

REFERENCES

- [1] A. Uchendu, T. Le, K. Shu, and D. Lee, “Authorship attribution for neural text generation,” in *Proc. EMNLP*, 2020, pp. 8384–8395.
- [2] A. Uchendu, Z. Ma, T. Le, R. Zhang, and D. Lee, “TURINGBENCH: A benchmark environment for Turing test in the age of neural text generation,” in *Findings of EMNLP*, 2021, pp. 2001–2016.
- [3] M. Sun, Y. Yin, Z. Xu, J. Z. Kolter, and Z. Liu, “Idiosyncrasies in large language models,” in *Proc. ICML*, 2025.
- [4] L. La Cava, D. Macko, R. Moro, I. Srba, and A. Tagarelli, “Authorship attribution in multilingual machine-generated texts,” in *Proc. ACL*, 2026, pp. 45 136–45 152.
- [5] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, “The state and fate of linguistic diversity and inclusion in the NLP world,” in *Proc. ACL*, 2020, pp. 6282–6293.
- [6] E. Stamatatos, “A survey of modern authorship attribution methods,” *J. Amer. Soc. Inf. Sci. Technol.*, vol. 60, no. 3, pp. 538–556, 2009.
- [7] U. Sapkota, S. Bethard, M. Montes, and T. Solorio, “Not all character n -grams are created equal: A study in authorship attribution,” in *Proc. NAACL-HLT*, 2015.
- [8] J. Nivre, M.-C. de Marneffe, F. Ginter, J. Hajič, C. D. Manning, S. Pyysalo, S. Schuster, F. Tyers, and D. Zeman, “Universal Dependencies v2: An evergrowing multilingual treebank collection,” in *Proc. LREC*, 2020, pp. 4034–4043.
- [9] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, “DetectGPT: Zero-shot machine-generated text detection using probability curvature,” in *Proc. ICML*, 2023.
- [10] D. Macko, R. Moro, A. Uchendu, J. Lucas, M. Yamashita, M. Pikuliak, I. Srba, T. Le, D. Lee, J. Simko, and M. Belikova, “MULTITuDE: Large-scale multilingual machine-generated text detection benchmark,” in *Proc. EMNLP*, 2023.
- [11] Y. Wang, A. Shelmanov, J. Mansurov, A. Tsvigun, V. Mikhailov, R. Xing, Z. Xie, J. Geng, G. Puccetti, E. Artemova *et al.*, “GenAI content detection task 1: English and multilingual machine-generated text detection: AI vs. human,” in *Proc. GenAIDetect Workshop at COLING*, 2025.
- [12] T. Kumarage and H. Liu, “Neural authorship attribution: Stylometric analysis on large language models,” in *Proc. IEEE CyberC*, 2023, pp. 51–54.
- [13] W. Zaitsu, M. Jin, S. Ishihara, S. Tsuge, and M. Inaba, “Stylometry can reveal artificial intelligence authorship, but humans struggle: A comparison of human and seven large language models in Japanese,” *PLoS ONE*, vol. 20, no. 10, p. e0335369, 2025.
- [14] K. Przystalski, J. K. Argasiński, I. Grabska-Gradzińska, and J. K. Ochab, “Stylometry recognizes human and LLM-generated texts in short samples,” *Expert Systems with Applications*, vol. 296, p. 129001, 2025.
- [15] B. Murauer and G. Specht, “DT-grams: Structured dependency grammar stylometry for cross-language authorship attribution,” in *Proc. GvDB, CEUR-WS Vol. 3075*, 2021.
- [16] B. Huang, C. Chen, and K. Shu, “Authorship attribution in the era of LLMs: Problems, methodologies, and challenges,” *ACM SIGKDD Explorations*, vol. 26, no. 2, pp. 21–43, 2024.
- [17] V. Padmakumar and H. He, “Does writing with language models reduce content diversity?” in *Proc. ICLR*, 2024.
- [18] Z. Sourati, A. S. Ziabari, and M. Dehghani, “The shrinking landscape of linguistic diversity in the age of large language models,” *Nature Human Behaviour*, 2026.
- [19] A. Panickssery, S. R. Bowman, and S. Feng, “LLM evaluators recognize and favor their own generations,” in *Proc. NeurIPS*, 2024.
- [20] A. Erlikhman, M. Tarekgn, and P. Juang, “Self-attribution is not self-recognition: A peer baseline for LLM judges,” Under review, 2026.
- [21] P. Qi, Y. Zhang, Y. Zhang, J. Bolton, and C. D. Manning, “Stanza: A Python natural language processing toolkit for many human languages,” in *Proc. ACL: System Demonstrations*, 2020.
- [22] M. A. Covington and J. D. McFall, “Cutting the Gordian knot: The moving-average type–token ratio (MATTR),” *J. Quant. Linguist.*, vol. 17, no. 2, pp. 94–100, 2010.
- [23] K.-I. Goh and A.-L. Barabási, “Burstiness and memory in complex systems,” *EPL*, vol. 81, no. 4, p. 48002, 2008.
- [24] X. Bai *et al.*, “Self-recognition in language models,” arXiv preprint, 2025.